

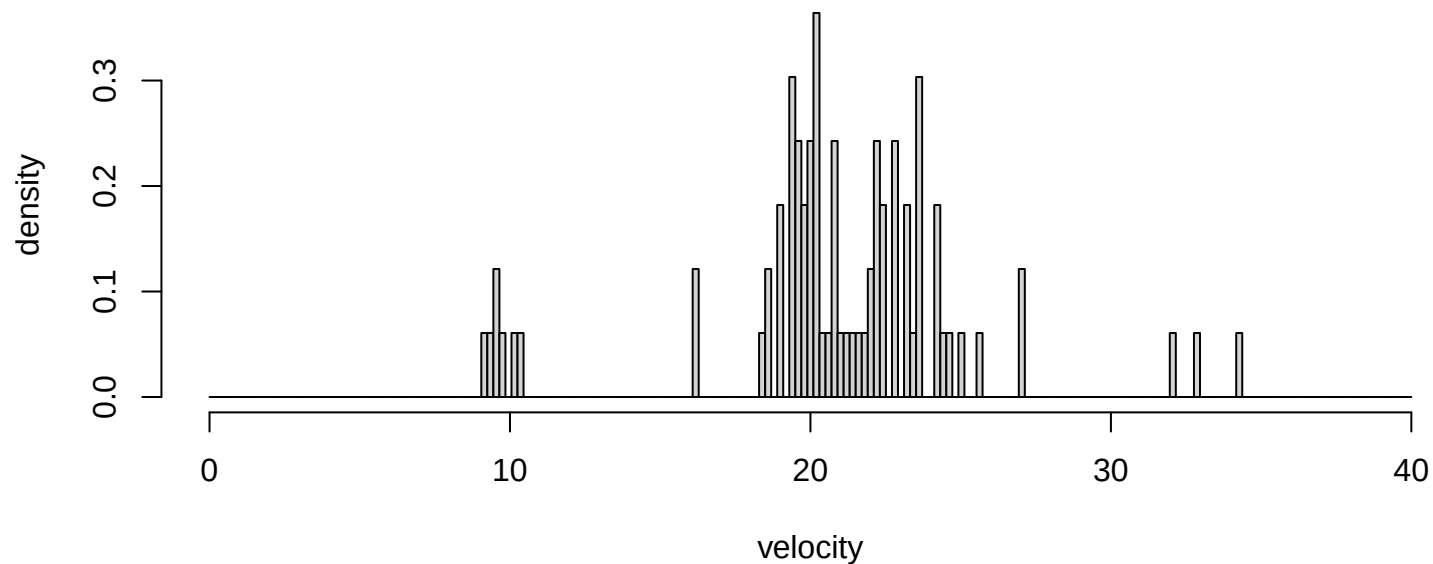
SC7 Bayes Methods MT25

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Lecture 13: Reversible Jump MCMC case study

RJ MCMC and fitting mixture models

The Galaxy radial velocity data are shown in the figure below. It is natural to model this via a mixture of normals. However we do not know the number of components in the mixture.



Likelihood:

Our data $y_i \in \mathbb{R}, i = 1, 2, \dots, n$ are independent samples from a mixture model with m components $N(\mu_j, \sigma_j^2)$, and mixture weights $w_j, j = 1, 2, \dots, m, w_j > 0, \sum_{j=1}^m w_j = 1$. Given $m \in 1, 2, 3, \dots$ the mixture parameters are

$$\mu = (\mu_1, \dots, \mu_m), \quad \sigma = (\sigma_1, \dots, \sigma_m), \quad w = (w_1, \dots, w_m).$$

The likelihood is

$$p(y|\mu, \sigma, w, m) = \prod_i \left[\sum_{j=1}^m w_j N(y_i; \mu_j, \sigma_j^2) \right].$$

If $\theta_j = (\mu_j, \sigma_j, w_j), j = 1, \dots, m$ then $\theta = \{\theta_1, \dots, \theta_m\}$ should be thought of as an unordered set, as any permutation gives the same likelihood.

Priors:

Model prior $\pi(m) = \text{Poisson}(m; \lambda | m > 0)$ with $\lambda = 10$ - centred at 10, and tails off above about 20 clusters.

For the parameter priors $\pi(\mu, \sigma, w | m)$, take

$$w \sim \text{Dirichlet}(\alpha 1_m) \quad \text{with } 1_m \text{ a vector of } m \text{ ones}$$

and take $\alpha = 1$ so w are uniform, summing to one,

$$\mu \sim N(\mu_0 1_m, v_0 I_m), \quad \text{with } \mu_0 = 20 \text{ and } v_0 = 10^2,$$

covering the data in $[0, 40]$ at 2σ - I assume the scale of the response is known *a priori* - and

$$\sigma_j \sim \text{Gamma}(1.5, 0.5), \quad \text{iid for } j = 1, 2, \dots, m,$$

again informed by the scale: mean equals 3; shape 1.5 rules out very dense clusters at small σ ; small rate gives heavy tail, standard deviation about 2.5.

Posterior: The posterior for the model and parameters (θ, m) ,

$\theta_j = (\mu_j, \sigma_j, w_j)$, $j = 1, \dots, m$, with $\theta = \{\theta_1, \dots, \theta_m\}$
and $m \in \{1, 2, 3, \dots\}$ is*

$$\begin{aligned}\pi(\theta, m|y) &\propto p(y|\theta, m)\pi(\theta|m)\pi(m) \\ &\propto p(y|\mu, \sigma, w, m) \\ &\quad \times \text{Dirichlet}(w; \alpha \mathbf{1}_m) \\ &\quad \times \prod_{j=1}^m N(\mu_j; \mu_0, v_0) \text{Gamma}(\sigma_j; 1.5, 0.5) \\ &\quad \times \text{Poisson}(m; \lambda) \times m!\end{aligned}$$

Note that for $m > 0$,

$$\text{Poisson}(m; \lambda | m > 0) \propto \text{Poisson}(m; \lambda),$$

as a function of m at fixed λ , so the condition can be dropped.

* θ is a set not a vector. For every set there are $m!$ vectors.

RJ MCMC algorithm targeting a normal mixture posterior

Suppose the state is $X_t = (\theta, m)$, $\theta = \{(\mu_i, \sigma_i, w_i)\}_{i=1}^m$. For irreducibility we need at least 2 variable dimension moves. I add 3 fixed dimension moves to improve mixing.

Step 1. Choose an update UAR, $\text{move} \sim U\{1, 2, \dots, 5\}$.

Step 2I. If $\text{move} = 1$ add a component (increase state dimension by three). We generate $\theta'_{m+1} = (\mu'_{m+1}, \sigma'_{m+1}, w'_{m+1})$ and set $\theta' = \theta \cup \theta'_{m+1}$ and $m' = m + 1$. Do this as follows:

Step 2Ia Simulate $\mu'_{m+1}, \sigma'_{m+1} \sim q_{\mu\sigma}(\mu'_{m+1}, \sigma'_{m+1})$ (use Normal-Gamma prior above). Set $\mu' = (\mu, \mu'_{m+1})$ and $\sigma' = (\sigma, \sigma'_{m+1})$.

Step 2Ib To complete $\theta'_{m+1} = (\mu'_{m+1}, \sigma'_{m+1}, w'_{m+1})$, we have to assign a value to w'_{m+1} . We have to maintain $\sum_j w'_j = 1$.

Choose a weight $j \sim U\{1, 2, \dots, m\}$ to “split”. Simulate $w'_{m+1} \sim U(0, w_j)$ and for $k = 1, 2, \dots, m + 1$ set

$$w'_k = \begin{cases} w_k & k = 1, \dots, m, k \neq j \\ w_k - w'_{m+1} & k = j \\ w'_{m+1} & k = m + 1 \end{cases}$$

The probability to propose m' given m is $\rho_{m,m'} = 1/5$. The probability distribution for (μ', σ', w', m') given (μ, σ, w, m) is

$$Q(\mu', \sigma', w', m' | \mu, \sigma, w, m) = \rho_{m,m'} q_{\mu\sigma}(\mu'_{m+1}, \sigma'_{m+1}) \times \frac{1}{m} \times \frac{1}{w_j}$$

exploiting the shortcut we mentioned in Lecture 12.

In the reverse move (move 2D below, decreasing dimension) pick component $i \in \{1, \dots, m'\}$ at random, delete it and add its weight w'_i to randomly chosen component $j \in \{1, \dots, m'\}$, $j \neq i$.

The probability (mass) to propose this exact reverse move back to (μ, σ, w, m) given (μ', σ', w', m') is

$$Q(\mu, \sigma, w, m | \mu', \sigma', w', m') = \rho_{m', m} \frac{1}{m(m+1)},$$

since we choose i from $m' = m + 1$ and j from $m' - 1 = m$.

Step 3I. Accept the proposal (μ', σ', w', m') with probability

$$\alpha^+ = \alpha(\mu', \sigma', w', m' | \mu, \sigma, w, m)$$

where

$$\alpha^+ = \min \left\{ 1, \frac{\pi(\mu', \sigma', w', m' | y) \rho_{m', m} \frac{1}{m(m+1)}}{\pi(\mu, \sigma, w, m | y) \rho_{m, m'} q_{\mu\sigma}(\mu'_{m+1}, \sigma'_{m+1}) \times \frac{1}{m} \times \frac{1}{w_j}} \right\}$$

Step 2D. If move = 2 delete a component. Set $m' = m - 1$ (if $m' = 0$, reject the move and set $X_{t+1} = X_t$).

Step 2Da Simulate $i \sim U\{1, 2, \dots, m\}$. Set $\mu' = \mu_{-i}$, $\sigma' = \sigma_{-i}$.

Step 2Db Simulate $j \sim U\{1, 2, \dots, i - 1, i + 1, \dots, m\}$, set $w' = w_{-i}$ and replace $w'_j \leftarrow w_j + w_i$.

The probability for the (whole) forward proposal is

$$Q(\mu', \sigma', w', m' | \mu, \sigma, w, m) = \rho_{m, m'} / m(m - 1),$$

and to reverse

$$Q(\mu, \sigma, w, m | \mu', \sigma', w', m') = \rho_{m', m} q_{\mu\sigma}(\mu_i, \sigma_i) \times \frac{1}{m - 1} \times \frac{1}{w_i + w_j}.$$

Step 3D. Accept the proposal (μ', σ', w', m') with probability

$$\alpha^- = \alpha(\mu', \sigma', w', m' | \mu, \sigma, w, m)$$

where, after cancelation,

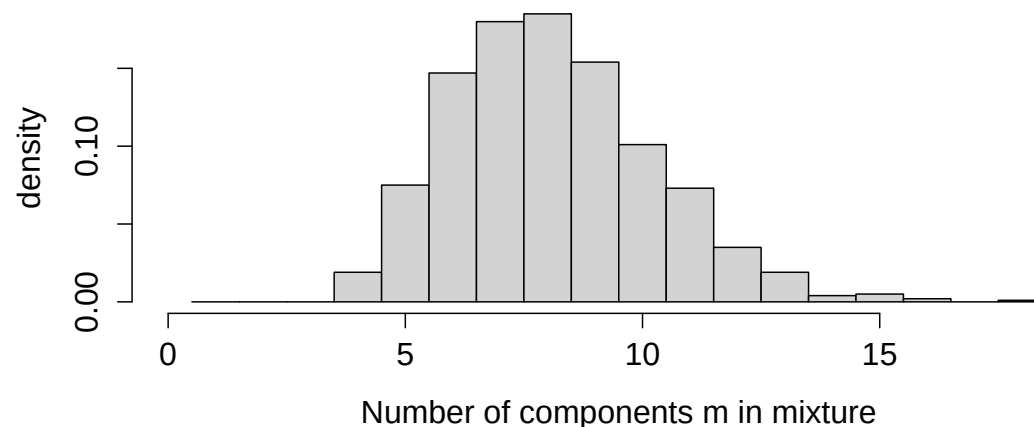
$$\begin{aligned} \alpha^- &= \min \left\{ 1, \frac{\pi(\mu', \sigma', w', m' | y) Q(\mu, \sigma, w, m | \mu', \sigma', w', m')}{\pi(\mu, \sigma, w, m | y) Q(\mu', \sigma', w', m' | \mu, \sigma, w, m)} \right\} \\ &= \min \left\{ 1, \frac{\pi(\mu', \sigma', w', m' | y) m q_{\mu\sigma}(\mu_i, \sigma_i)}{\pi(\mu, \sigma, w, m | y) (w_i + w_j)} \right\} \end{aligned}$$

We have additionally moves 3-5 which act on μ , σ and w respectively in fixed dimension moves.

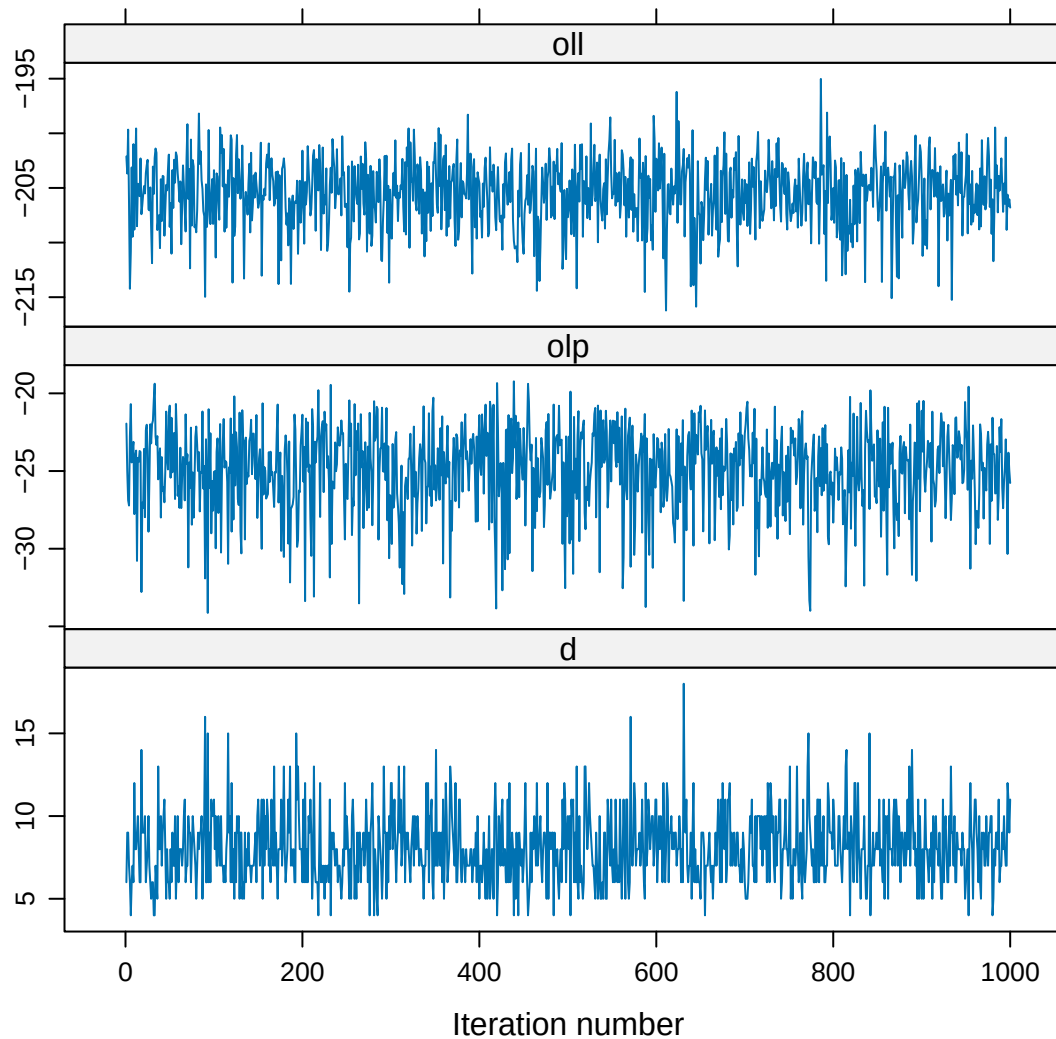
The code was checked by setting $p(y|\theta, m) = 1$ and checking the MCMC output was distributed according to the prior.

RJ-MCMC for a normal mixture

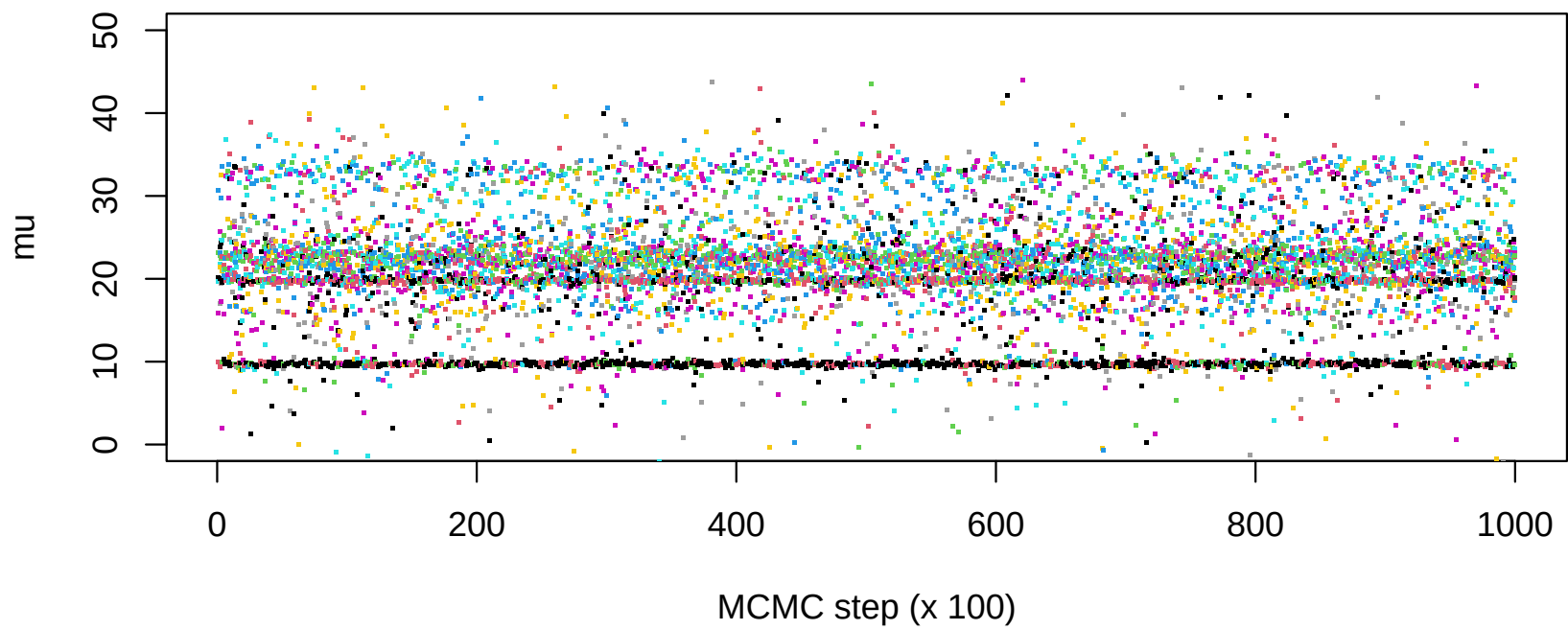
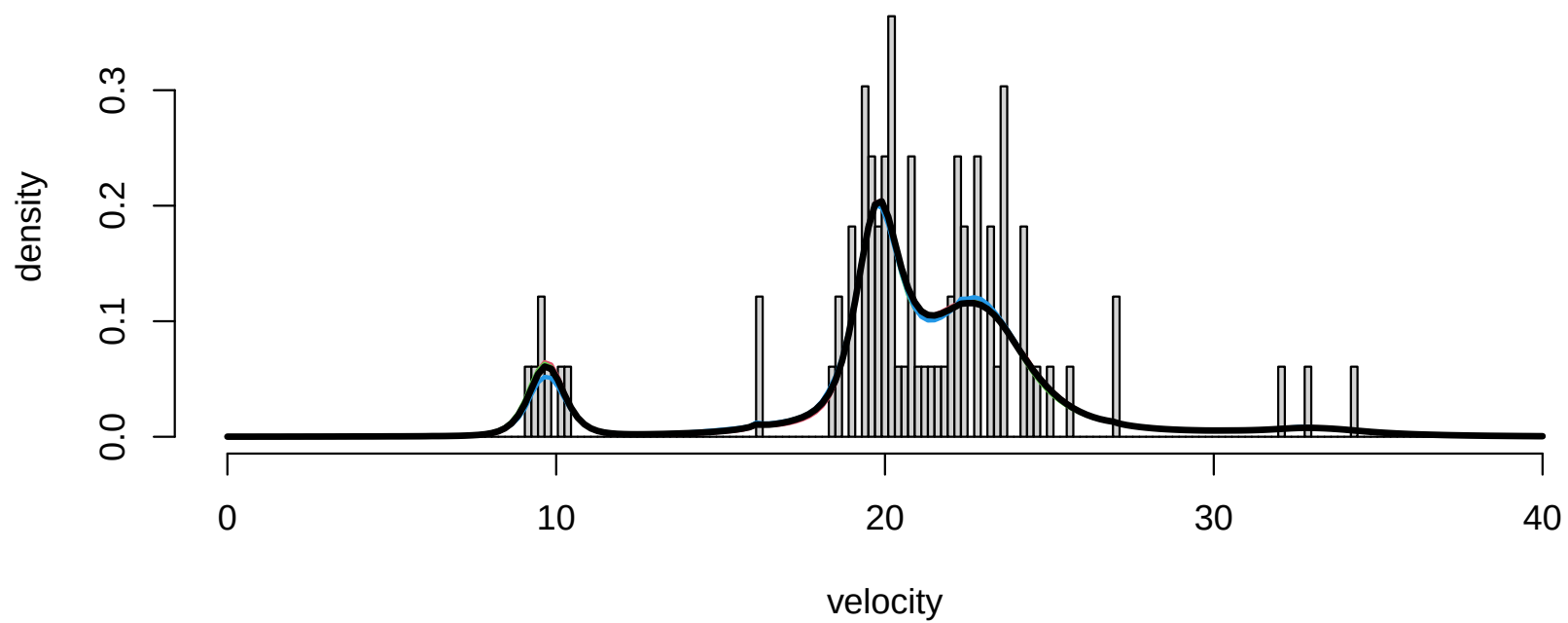
We illustrate the method on the Galaxy velocity distribution data. The R-code and further detail of the algorithm are available on the course website. We ran the code and generated samples $(\mu^{(t)}, \sigma^{(t)}, w^{(t)}, m^{(t)}), t = 1, 2, \dots, T$ from the joint posterior distribution over the number of clusters and the cluster weights and parameters.



The plot above shows the posterior distribution over the number of components. 6-9 components is the number favored.



The plot shows traces for the log-prior, log-likelihood and number of components, (as the number of parameters vary, they are not easily plotted).



The bottom figure shows the sampled means $\mu^{(t)}, t = 1, \dots, T$. The dimension varies with t . Point $(t, \mu^{(t)})_i$ has color i (default R colors). We can see label-switching. Stable mixture components (the horizontal bands) have varying label, so varying color.

The top figure shows an estimate $\widehat{p(y'|y)}$ of the posterior predictive distribution $p(y'|y)$ (black line) at each point y' on the x -axis. Since

$$p(y'|y) = \sum_m \int p(y'|\theta, m) \pi(\theta, m|y) d\theta$$

we use the natural estimate

$$\widehat{p(y'|y)} = \frac{1}{T} \sum_{t=1}^T p(y'|\theta^{(t)}, m^{(t)}).$$

The underlying histogram in black is a histogram of the data, y . We expect the distribution of the data to match the posterior predictive distribution, and the fit seems reasonable. Other lines are $p(y'|y, m)$ -estimates ($m=4$ red, $m=6$ green, $m=8$ blue).